

ActInf ArtStream 002.1 ~Aenne Brielmann: "Modelling individual aesthetic judgements over time"

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all right hello and welcome everyone it is July 8th 2024 and we're in active inference art stream number 2.1 a computational model of aesthetic value with anman and we'll have a presentation followed by a discussion so thank you very much for joining to you for the presentation and looking forward to it um thanks so much for having me Daniel I'm really really thrilled to talk about a computational model of something that a lot of people still doubt that machine models can ever attempt to model um namely aesthetic value and that's been the work I've been doing for a couple of years and today I don't necessarily want to talk so much about the concrete content of this model which I do a lot with my papers and it's just as important I think what's also important and often gets left out a bit from the conversation is the question of why we should even bother um developing and applying a computational model to something so complex as aesthetic valuation right so this is going to be a really high level overview and I'd be really stoked if we could have the discussion about this because that's really what I'm trying to instigate here so stuff that I want to talk about is that it Valu is grandiose words with a rather intuitive but also quite fuzzy meaning what I'm talking about personally when I talk about aesthetic value is something actually quite mundane it's the pleasure that we derive from any sort of sensory experiences and you all have felt that value and acted according to it when you decided to go to a concert or listen to song when you went out into the mountains or any sort of landscape and just gazed at it um when you went out to a restaurant and decided what you want to eat and savored that taste or even when you touched something or someone and it just felt so good you wanted to touch to linger all these experiences of pleasure that are based on the sensory properties of the object to interact with I personally call aesthetic value and my mission for a couple of years has now been to find a computational model that can tell us something about how these values emerge and what they do to our Behavior to answer that question the most important foundation to me personally is

to ask the inherently computational question um if we take it to the original meaning of the computational of what this so-called aesthetic value is there for what does it do why do we have an emotional effective reaction towards sensory properties and in the past people have come up with two very different families of theories to explain these Origins um one of them um I will in short D fluency theories are very much about what I would call Lazy brain saying that aesthetic value actually indicates to us that we are easily able to process the sensory stimuli and our brain likes that because it needs to do less work in a sense to simplify horribly um and examples where that has shown up in the data is when we look at symmetric stimuli so only on half as we process or when we look at the effects that we get from familiar or exposure so the more often we see something the more we tend to like it and that might according to fluency be because we get to be able to process it more easily becoming more familiar with it and this is very much an evaluation of the present moment how good am I right now in dealing with my sensory environment this theory has very often been criticized for being overly simplistic and not very much looking into the future because it's so grounded in the present moment and so in came a different family of theories that I will summarize under an umbrella term of learning theories um that very much rest on the assumption that aesthetic value is actually a signal for learning so not so much present moment adaptation but rather a look into the future gauging how well the current sensory environment can serve me in learning about the future sensory environment as well this is all about making inferences about recognizing errors in our perceptions and then reducing them so the value here comes from a reduction of processing errors of errors in making predictions about sensory environment um and in that sense we striving to adapt for better adaptation into the future so both theories have one big thing in common and that is the underlying goal here is to make sensory perception processing good with different emphasis on the present moment in terms of fluency and the long-term future when comes to Le when I discussed the theory with my D supervisor Peter Dean we were both kind of struck by the simple question about these theories Would we not want to have both would we not want to simultaneously be good at processing your world as it currently is as well as learn about potential future so the model that we came up with wants to achieve both goals concurrently um and has to weigh off between the two so the sensory system striving to both recognize what is in front of it right now but also wanting to predict as best as possible what is going to encounter in the future um and you see that in everyday life too right so if you want to work with your sensory system just crossing the street you want to both now be able to recognize what is in front of you do I know this person should I greet

them but you also want to be able to make accurate predictions about whether or not that car is going to hit you the next second if you do proceed to cross the street and this is a very very prominent theme throughout most models of machine learning um that are based on theas of reward learning there is an agent that right now decides its next action based on what is rewarding in this very moment willst taking into consideration whether its predictions and the action that resulted from that led to better RSE outcomes than expected so predicting the future evaluating it right now and then adjust Behavior accordingly and what we I I believe is why should the sensory system and the value that comes from that this aesthetic value be any different from those with that thought in mind we proceeded on um building that but what does it look like so just taking one step back for those people here who have not kind of sat in front of their computers and actually done something with these things this is what a computational model looks like it's it is just a bunch of code it's a bunch of mathematical formulas realized in any programming language you like um and you can run it on simplified comput computer representations numbers of the things you want to simulate and derive predictions from it um and this particular one for athetic value is implemented and you can download it from GitHub and I very much try to keep that updated what is key for its usefulness for people like me and I'm a psychologist by training is that these Expressions the definitions and the terms using them um are interpretable right so every function here um serves a as a proposed mechanism Um this can be very high level but it can in other cases go down to the functions of individual neurons that are being modeled here um most often though we see this interpretability in the parameters that are passed onto these functions so what are the settings under which these mechanisms operate and that is true in this particular model as well what our model does to gain this interpretability is that it has two sets of functions that interact with each other two parts of the model that do roughly represent this one famar theories of processing fluence and it also has one that represents the mechanism that um represents learning from prediction errors that gives reward for prediction error reduction so this looks like in a sorry one second there was my speakers going up um my headphones going out um and we can repres Us in in a comic like way right where we can actually at least in a simplified one-dimensional structure represent what the model looks like so it is nothing more or less than um the representation of the expectations of the inner system so a probability distribution of what the system expects to see right here right now and it takes a readout of that probability so what are my predictions in terms of what we see here is we look at a dog one relevant feature of the dog is how long is its fur we have an expectation of how long it is we

read out how well the current environment meets our expectations and that read out is one component um so this is one of the functions that lets us know um the reward that we get out of here the value from immediate processing fluency right now and we can do a similar thing um illustrating how learning Works in this model how we actually bring in the first component of learning by shifting our expectations um as you see here by shift in the distribution according to what's present right now so I see a dog with long fur hence I expect dogs to have longer fur um also expressed as a function and we can go on and then say okay how do I represent errors how can my current state of beliefs be in contradiction to an expected future and so we can simulate that by saying we have a mechanism that evaluates how different is my current expectation to my expectations of the future and we can then over time evaluate how these long-term expectations match up to the current learning process and whether the learning in by itself gets us closer to long-term expectations or removes us further from them and that term then gives us what we call a reward of learning is the development of the prediction error over time um and that I horribly gloss over and I'm super happy to give you more details but I really wanted to get to this one very important Point like telling OKAY computational model is nothing fancy what is important is the idea that stands behind it and how we real ize it but as just as important to me as a researcher who very much does Empirical work is ideas are good and well but does it do anything and that's what I was very happy to work on with my master students where we put took the model implemented in computer code and said can it actually produce something produce predictions that are anything like the behavior we see for people people um and here we looked at one particular finding first with simulations that we know very well very many Studies have looked at it it's called A Mir exposure effect and it's depicted here on the left hand side and it basically looks at how much people like or enjoy something on the Y AIS as a function of how often they've experienced it and what we usually see is an inverted U-shaped curve where up to a certain point more exposure more familiarity leads to more liking but eventually this trend is going to turn on itself and we get bored and our model as you can see from the overlapping lines um comparing the data to the model's prediction is indeed able to replicate these findings and its settings so its parameters and how they're expressed um can tell us something about how this effect came about and again glossing over the details here very happy to talk about it later but it basically gives us a constellation um a geometry of settings that allow for this effect to occur and much very importantly discards all other options um and most um surprisingly to me what a model could do very well is in a completely separate experience experiment with our own data it was able

to predict individual people's judgment on how much they liked images um very well on an individual trial by trial level so we could say for individual trials that the model never saw in the sequence what a participant would grade them solely based on their other responses um so the model was indeed able to mimic something that was able to produce the same responses as humans would do and again Our Hope here is not to say oh great now we can build the best you know recommendation algorithm for any um internet platform you could imagine based on such a model um what we are excited about is that the parameter settings behind us tell us something about where these judgments actually came from um and this model performs better than other more naive model so what you see here is a comparison um on the y- axis of the error that our model makes versus on the x axis something that um is very often so used in Psychology namely the assumption that everybody responds like everybody else on average and that always makes way greater um errors in predicting people's responses than when we take a more complex but individualized approach to things and that's very reassuring for us um yes so it does do more than these more naive methods in the sense that the difference in the parameters suggest reasons for differences so what is so very often said about how we um do our evaluations of aesthetic experiences that they're in the eye of the beholder there's now something to a back up that these are not coincidental and be something that suggests where they might be coming from and I think there's even more we can do with this kind of approach to even a subject as pzy and sometimes hard to study as aesthetic valuation right so we can test different variations of this model against each other we can leion the model right so we're not hurting anyone here we're just removing parts of the code um and see which elements of the model are indeed essential to its functioning um and again find out more about the possible mechanisms that are indeed involved and crucial to this process um we can and this is to me exciting as somebody um who does not come from the computer science world uh or the machine learning world now compare this model to what I mentioned in the very beginning the same architectures were employed for decades to teach machines how to learn and um to understand why rats exhibits um certain behaviors right it's the same architecture what does that imply about aesthetic valuation as a reward learning model um and because it is so similar to the reward learning models that underly um Machine Vision um and a lot of the deep convolutional neural Nets we're using to process sensory stimuli most often visual to translate them into machine language we have a rationale to use these highly potent image processing architectures and use them to quantify what is in those images also in our research um and it offers the possibility to simulate right so we can now make predictions before we even start an

experiment about what people should do if our models were correct if we have one like this um and I will end this here for this particular presentation um and I'd be super stoked to start a conversation with you all right awesome cool okay lots of ways to uh go I'm looking forward to discussing it and if anyone watching has a question then they can type it and I'll ask it um let's just start with how did you come to be studying this intersection of questions and methods and and theoretical perspectives give me one second to put out my alarm that has been started ringing in the background um it's an amazing question so as I think for many people my journey started a little bit coincidental so I was initially very interested going to arts school but was very much dissolution so I've always been very interested but diathetics in the broadest term I play instruments I paint I draw um and had a supervisor who very randomly asked me whether I would want to study that but as a psychology PhD and that supervisor already was in a place both in terms of colleagues as well as his own personal history um that bound him very closely to computational methods so I studied at NYU with Dennis P who is a mathematician by training and the psychology department there is very fond of very quantitative um methods and building their own models to explain how the world works and that's how my journey started and this perspective has never been forced on me but it's the one that I was brought up with and I always found it very very useful for the reasons that I just said um in terms of being able to integrate a field that is often a bit separated right there's a lot of people by now doing research on aesthetic experiences but often very removed from the rest of the world um and I always thought well why not talk to the people we are talking about visually appreciating things so it's got to do something with how the entire visual apparatus works right so why not use the methods of those people who do the more cognitive or quote unquote basic side of things and apply them to this more effective evaluative side of things um and then and and via that I became really interested in computational modeling because I started realizing that experiencing beauty or pleasure from something is not just something um that is a nice you know it's it's not icing on the cake off life it's very ingrained in everything we do really especially in the industrialized Western World so I became very interested in this question how does this influence our decision making why do we you know spend a couple of hundred dollars uh and a lot of time and effort to travel somewhere to watch somebody play music for two hours right it doesn't make any sense and then I went to tubing worked with Peter Deon who is somebody who's worked on reward learning models um compute very deeply computational models for very many say and I came to him and said I have a hunch that these models apply to what I've been doing for five years too are you willing to work with

me on that and luckily he said yes cool thank you yeah it's interesting that Aesthetics seems to sit in a way between perception and behavior and also philosophy in that it's often seen as a philosophical area however one that deals quite explicitly with the perceptual system that might be addressed by psychology otherwise um okay you mentioned interpretability so that was in the context of the computational modeling H how do you think about interpretability of aesthetic models when we have this kind of dual interpretability from the inside like our own interpretability of the underlying phenomena and then we have this kind of external interpretability whether that's observing somebody having aesthetic experience or doing modeling like how far can we get with interpretability and what does that look like here yeah that was a really good question so I think probably the furthest we can go and aesthetics that might be a long way off is you know linking model components to physiological biological reality and being concretely able to say this function represents one neuron this function represents you know how signals are transmitted along the axon and so on and so forth right they really pinpoint every bit of the model to something that is both functionally important and physically observable um but to me interpretability in the state a Statics at is the best when each parameter in the model relates to something that is part of the theory that underlies it um so that when the parameter changes or becomes irrelevant that is probably the most crucial one we have justifiable reasons to change our Theory or adjust it right so what in our model for instance we have two parameters that weigh the two components um against each other so one parameter scales how important the um value is that is derived from this immediate um fluency signal and the other one scales how much value we derive from the um prediction error reduction from the reward of learning right and so if we see consistently over time that we can set one or the other to zero and nothing much happens that very clearly indicates that one component of the models is frankly Obsolete and doesn't play a huge role and that to me is very valuable interpretability in that sense interesting so we saw the inverted U exposure curve are there phenomena that kind of buck that Trend or are there maybe pieces or genres where we have like a step like we enter into it and then we can see more or different like how do we account for not just Rel listening 25 times to the song but you know 2500 times to the song yeah yeah it's one of my favorite things to ponder about as well um so there are different ways that I've discussed it with uh different colleagues and I believe that we someday might you know break it down to our internal models just having slightly different settings but perhaps the easiest explanation for it is that there's some stimuli that just sit so very firmly at the center of our expected future beliefs right so that there's basically never a

negative learning experience from them they are the Prototype of what for this type of stimuli at least we expect the world to be and remain forever more um and so that then leads the model to predict not an inverted U shape uh but an ASM toote so you basically stick around with very high Pleasures throughout and it never goes down um that's the easiest way of explaining it the more um everyday explanation if if you will one that we cannot test as easily in a lab setting is that um we never listen to the same song 25,000 times in a row without anything else getting in the way right and if that is the case you know we can always kind of swing back with our expectations and then come come back to it kind of with a reset so it's not that it's continuous exposure to the same thing all over again but each time you come back to that song you've changed a little bit your expectations have changed a little bit and so you're learning a new and if it sits at the right place of your expectations about the world you're always learning something good and so you continue this infinite Loop Vias short intermission of being exposed to things that are you know less enjoyable make you unlearn something that was actually beneficial H that's very interesting it's like a Smelling Coffee cleansing the palet and then relearning oh that I like that food but also I'm kind of relearning it um I I was also es especially when when you connect the fluency theory of Aesthetics to the expectations I was struck that there's the expected free energy construct for decision-making and it has this pragmatic value element of preferences expectations and then the epistemic value with the learning and so initially I was wondering well aesthetic value which is what motivated the um Talk would Aesthetics be a third elements in the consideration um or it's it's almost more interesting that it exists across these two areas of expectations and that has more of an archetype of being like well it's this time so we hear this song so that's the kind of like pragmatic side of Aesthetics with a cultural scaffolding of expectations and then there's also this learning intrinsic novelty element so it kind of comes down just like action selection in the general case it comes down to these kind of two complementary but but non-convertible types of of impact that the stimuli can have yeah I think you exactly captured the intuition of that um it is debatable whether that is all the there is twoips and I personally do not think so I think this model does a very good job at capturing these two aspects um but you know the more I think about it of course the more incomplete the model becomes um so yes I think there are these two archetypes of also aesthetic enjoyment I've also seen that across other completely other theories like not not psychological theories of Aesthetics right there's this kind of um Cinema pop culture entertainment aspect of oh this is easy and comfortable um and then the more challenging you know Fine Arts aspect of Aesthetics and I do see both of these discussed um

both jointly and independently uh what I think in terms of selfism although the model is still lacking is um maybe that's what you meant with aesthetic is a contextualization in terms of the sensory system does not exist in isolation in our brains it is also um coupled with everything else so elements of nostalgia um of Association of of memories that is something the model completely disregards it just doesn't have it baked into it um and so it's completely and entirely possible we're missing an important ingredient St but again you know you cannot only leion mod you can also amend it you can add another component to it so if anybody's up to could be very thrilled cool yeah looking for those like residuals or looking at what factors matter where the model doesn't predict well I I was I was definitely thinking about um in light of that sort of pop art and decoration on through more like epistemically challenging art sometimes the they they what changes isn't necessarily the medium but often is sort of like the contextualization like the music mum setting you're about to be walking into this room where this is going to be there and then the object as the sort of um recent Decades of of art have have shown like that object can be anything and nothing and so then that is like wasn't this about perception but then what is there to perceive in this blank wall yet also that becomes kind of like the exception that proves the rule because it it starts to tap into the more contextual elements and and possibly reveal greater differences amongst people rather than something that that like um a ratio that strikes many as pleasant due to deeper priors about shape yeah yeah no definitely and you know you you can try and fit that into the model too you know you can go ahead and that's what we we maintain where it might continue to be useful is that um context might bring about a drastic shift in parameter settings right so as soon as I go into the museum I'm no longer looking for something that is immediately easy to process right now I'm voluntarily getting myself into a context where I know that there's more value to be found in learning and challenging myself and so that might be one way of saying okay that's how you know we can think of the model and not only modeling differences between individuals because that's how it does that too for instance but also between contexts um that being said there's also completely different shifts that might be more difficult to explain with that um my most recent example is um it's reping me as a bit of a nerd but the author of The Dragon Ball series recently died and I think along with a lot of other people I Revisited something from my childhood is without that lens added to it wasn't very good art to begin with but got a different kind of appreciation for it with like an whole back layer of appreciation and historical importance and whether that is just parameter shifts that happen kind of snap in an instant I don't know so maybe we're still missing something yes Nostalgia and memory and a given

stimuli having like a pragmatic value simply because it was learned to be expected and so then it all things being equal is more likely to be part of the expectations going forward just part of the historical learning um okay so you mentioned um probably a a a illusion that that must be included in every such talk about beauty in the eye of the beholder and yet what is in the eye of the beholder is in the mind of the beholder with their generative model so H how would you say like predictive processing predictive coding active inference how are these cognitive theories stepping into Aesthetics in the modern era and like helping reform or or reinforce some of these more ancient sentiments on perception I think in recent times um predictive coding has very much reinforced first the notion of Beauty and the eye of the beholder because it in a sense offers a very easy explanation to so as as to why there is so much subjective variability with the notion that you know objectively what the world gives to us in terms of physical stimulation is not what everybody experiences right it's an active constructive process perception is not something passive that just happens and so for everybody it creates something different so it should be logical then that we also appreciate things differently however I think it's a very healthy development in that with predictive coding much as with other models including all the computational models it's not this story of hopeless oh well we everybody's different so what we're going to do about it now we can't explain it possibly it offers very good theoretical foundation for understanding differences not as a nuisance or measurement error but as something that's that's actually at the heart of the process itself right um I think that's quite promising and I think also in that sense that kind of scientific approach to Aesthetics receive much more um positive feedback and welcoming from the more artistic Community because it does not want to take away from differences does not want to explain them away as Nuance but very much takes them seriously as um something that is a Justified result of the underly processes thank you yeah that's very interesting like the account of the painting is not seeking to explain away the differences in shading or color the account of different people looking at the painting also would not want to explain those away but rather like account for the distributions as they are and continue the investigation um so you mentioned algorithms how would you say your work interfaces with like algorithms and recommendations especially in a era not just of recommendation of extance content but generative and synthetic methods I mean personally this model represents pretty much everything that these algorithms want to do right it it does not have a generative part yet but it's architecture makes it blatantly obvious how the generated part should be constructed right you can construct a um General adversarial Network out of this no problem you could train it

um in a tener model with what we have right now and you know if you have enough data sure you could do it the only thing that's not feasible about it is you would have to have complete surveillance over people and a good way of getting continuous output about their reactions and then you you could have um every platform stream you know kind of continuously predicting what's that person going to like best next um and that's kind of scary in in that sense right um but I also think it's hopeful in the sense that what current algorithms are doing is really not taking the to any consideration at all so at the heart of the model as we have it lies learning and lies the fact that we do not want to be stagnant right we do not want the same experiences over and over and over again we do get bored the exposure effect does eventually turn negative um and in that sense offers also always due to it not being this one comic I showed but multi-dimensional always many options of what you know would as next experience be rewarding and be valuable to you um and in in that sense I always want to kind of build the model out in that way because it would get away from the current State of the art which is very much cluster driven putting people into okay you're this part of the cluster you know okay so you like rock so you probably like hard rock and maybe country rock let's try that okay you didn't like that maybe you're more of a hard rock person and we keep you firmly um and only recommend you artists in that corner instead of making a jump um so yeah I think everybody who works with artificial intelligence in whichever capacity these days kind of has to ask this question right what is what is the benefit what's the worst case scenario and and in what way could it be helpful and that's the I could come up with so far it's it is a dark but fascinating like aesthetic pen Opticon with the biomarkers and the processing happening faster than the time scales of our own aesthetic judgments and then like updating with recommendations or with tweaking and modifying in in in ways that that we don't necessarily even see but then it's like but what if that journey is taking us on these epistemic flights and what if the learning rate is being calibrated to a kind of sensation um that couldn't be provided like by people making arts for each other and I wonder if that will be one of the Technologies or or kind of manifestations that that will really clearly highlight that contextual element because very Salient stimuli will be arising from that kind of a process and then also there's going to be like the drawing that my friend made for me on the refrigerator which is like static but then it's going to have this embeddedness and the connection and and then maybe that's going to still be a thing to look at yeah absolutely I think this is also I'm not a doomsday Sayer when it comes to um this ties into creativity a lot in my opinion right so what can we gain from generative art in any way sense and I not want to say it's all doom and gloom and it's all bad um but what you just mentioned

is kind of embedding of a work into a personal history be it with the viewer or with the artist that we as viewers can then learn about is something that you know kind of goes beyond the scope of what we're doing um these days where you know our modeling is very much reliant on quantifying the objects and then trying to infer what is going on inside the view we're not quite there yet where you know we also kind of infer the underlying non-physical properties of the stimulus itself beyond what is provided by the viewer as well but maybe one day in the future you know we're going to have it solved and Aesthetics is going to be Trivially solved asid um H H how would you represent the um the the philosophical or psychological or computational developments on art recognition models in contrast with models of creating art or different Medias and different ways of more actively constructing art it's a really good question I think they're approached right now from two very different um groups of people um so everything I think that we see these days in terms of um generative approaches comes from um and and I say that as a daughter of an engineer um comes from Engineers people who are very product focused right who want to give have an output that can be practically used for something and as such a lot of by now very impressive Black Box models have emerged that can produce things that are very impressive like 20 years ago nobody would have guessed that we could would be able to do this but without any real understanding of how that emerged from you know tens of thousands of example images that it was trained on on the other hand these people have become less interested something that is also still being done um that is modeling people's responses um so these algorithms are mostly really not good because they don't have enough data and not good not high enough quality data um the best data sets you can get for training a model to um on a large machine scale model is from like an old page from a photo contests um there like five or so votes per image which for psychology psychologist sounds like horrible horrible horrible sample size the horribly biased sample um and so these efforts I think by now have to stand back to what is now more coming from the um cognitive Sciences from psychology Neuroscience um and those models are very different because their predictive quality is often not quite as good like they're not as correct as [Music] um models would you know 10,000 of times of parameters but they're very interpretable um so I think there are two very different worlds that also want to do very different thing one wants to deliver a product and the other one wants to deliver more of an explanation that's making oh go ahead yeah and and the sad thing is I would very much love to see them you know come together but I don't see that quite happening yet H that that's making me think about um again with interpretability and the um our own observation of observing art or

creating art and what factors we feel are associated with our own perceptual experiences and our own generative experiences like a funny quotation from William Blake he said the difference between a bad artist and a good one is the bad artist seems to copy a great deal the good one really does copy a great deal and yet then I hear that the kind of synthetic methods that we have today are like glorified autocomplete or copying which kind of even scales back the conversation to like but then would that still be art and potentially even more artlike or would that somehow make it less art like and that conversation about what makes something aesthetic what makes something decorative versus high art how are these longer term Trends in what is perceived or or discussed as art that is just like instantly we're just like back in the whole fabric of of society and the complexity and the uh uh and the challenge of coming up with a neat account of like any phenomena yeah absolutely and it feels like we're back in you know 18 19th century with that really reminds me this entire debate that we're having right now you know is this art oh my God what is happening to Art and artists at all it reminds me very much of T when photography was invented in a way um it feel feels like de is U repeating itself in in variations which also art loves to do right repetition in variation um and so I guess I took the easy way out with that in my Approach right really saying I'm not even interested if you call it art I'm also not very interested if you call it aesthetic um very much interested in what you feel when you experience that and what that feeling makes you do and then I I Love My Philosophy colleagues I really do I love working with them but I leave it absolutely up to them to discuss defin semantics and terminology of when exactly that crosses the border from pretty to beautiful to aesthetic to transformative I'm perfectly F sitting here saying I I can at least predict that they're going to they don't like it h how how do you think about that like reported unreported revealed question like is there another kind of information or way that you look at what people say in light of other data yeah I mean the way that I interpret your question is you know what do we do with this play of psychology when it comes to these very subjective experiences of you know how do we interpret what we're getting in terms of data right that where getting at um so my ideal that I strive towards and I do have data that is just not good enough yet right didn't think it through a th% is looking at people's behavior and make inferences from these behaviors because um if we only observe for instance the setup was like this um how long people will look at a given image before they decide that they want to go on to the next one they do not really need to think about what we want them to say and they don't really need to think about oh what's going to tell you about me that I say I like this image very much or not um because there's concrete behavioral

consequences for them in the right here and right now if they display a certain action so you know um I can give them a fine art piece uh and if I ask them to rate this and they recognize it and they're like oh yeah okay this is a museum I should probably like this otherwise I'm horribly uncultured it's very easy to do that right there's no negative consequences for them to experience whether they Mark a one or nine on a 1 to 10 scale who cares but if I say listen I'm going to leave you here for 10 minutes um and the only thing you're allowed to do is look through these images um then there's the real consequence of looking at this fine art piece and you know deciding oh maybe should I really look I really really don't like looking at this but I should like it right and that's probably a better method but it's also harder to pull off right so I do think behavior is harder to set up but if set up right the better more direct measure of what's actually going on but for all intents and purposes self-report as long as you do not give people a reason to want to distort their reports is probably the most direct and easiest to do um and we are really lucky in Aesthetics right like we do these kinds of experiments mostly with stuff that people are very happy to do right we all love to talk about our favorite music and our favorite movies and we like to give our opinions about these things they're important to us so we can do self-report easily it's just not perfect never going to be interesting also that that like in a way recapitulates the kind of epistemic foraging or optimal foraging type experiments like there's 100 trees they have different quality and then you could just from the movement patterns eventually parameterize a model like to what extent was the switching Behavior driven by saturation of learning versus saturation of of the preferences and then here it's again that aesthetic value has elements and then your model is like like a Cornerstone for adding in some other considerations but um at minimum it fuses those two threads of theory and gives the kind of minimal playground to to start to explore like how those work and and also how those might work in Minds very different than humans like looking at the response curves for training neural networks absolutely and Bo oh boy have I been sitting through like I can't tell you to how many workshops and stuff I've been going on foraging behavior and trying to do that um and and then I got hung up on boredom that's how it sometimes goes right and then then we discovered this thing about getting bored about things and interestingly boredom wasn't boring at all to think about and so I got sidetracked but I think what you said right about now is it also so cool to start thinking about and again just making these connections um with foraging Behavior because honestly when I pick up my my phone and decide what music I want to listen to for my bus ride to the office that's kind of exactly what's going on I have this limited time window to get hopefully maximum enjoyment out of um and it's exact just like do

do I keep digging in this corner of songs or like is it more rewarding now to kind of start something new and it's exactly that that problem so yeah that's yeah yeah it could be people in the museum like the physical location the diffusion and then there could be like um different I mean it's it's like a resource location and um but it's very complex there's there might be like a more visible so then that's where like that sort of like social thumb on the scale of Aesthetics but then maybe somebody's Aesthetics concept includes like performativity at which point then there's it's not it's not a um illicit ad mixture to have the social value like standup comedy or some other kind of performativity where having other people's gaze is part of what is being constructed whereas in other kinds of art there's like I mean even evidenced by earlier in our um discussion like there's a feeling like we could isolate it to the viewer or to the experiencer but then that is almost shown to be kind of like a contrived case not the natural history of how the art is made and used yeah absolutely and that that kind of I think it's also goal that other people are working towards right now what you mentioned is that these single agent models that we all very much rely on these days is also not telling the entire truth of the experience right um it not just my this entire social component as well if we're talking about these kinds of real life settings and experiences that also bring in their F to the completed picture yeah um okay in the last little bit if anyone wants to write a question they can otherwise H how do you see your work and research and everything unfolding or what ways are you excited to go same my dream for all of this really is to get a lot of people excited and trying out these kinds of models um of course for me personally most exciting thing would be people picking up what I've been doing and tweaking it and really thoroughly testing it and figuring out what is wrong with this model because I'm sure there's something wrong and getting to a very solid core understanding of how it all sort of relates today I've painted very much a picture of what has um basic visual but also other perception to do with the valuation of these experiences but that can be tied up so much further into how do these EV these valuations then impact our behaviors how do these behaviors lead to changes of experiences um and really take seriously this idea that there is no passivity in the perception of our world it's all a very much active process that keeps on changing um and I think we can get a good understanding of that and I think we're very exciting times um especially for Aesthetics to build these models in such a way not only that they can be useful tools but really give us a good understanding of what aesthetic experience truly is so I hope to be able to continue working on that for very many years to come awesome that idea of no passivity kind of period is a big theme in active inference and it it makes me revisit the beauty in the eye of the

beholder and about the is secting and about how with with gay tracking the dwelling could be associated with our like a local Maxima looking at at a interesting feature of an architecture or a specific location and then the Cade pattern which few if any people can directly control is driven by epistemic gain and that's this kind of like hierarchical locking in like there been a lot of experiments in active inference facial recognition and faces that are the right side up and upside down and the kind of epistemic trajectories that are taken so maybe that's one way that um predictive processing also like revisits and enriches the beauty in the eye of the beholder because it it brings action into that eye instead of just to to be hold as a passive process that would be like well you know the images in the lens of the photographer would be a passive stance but then even even the I which is kind of like a symbol or has a connotation of of receptivity but of course its action is is so active and and ceaseless even on a simple scene and then maybe even um mysterious art like a solid color square that then becomes like this playground for The cading and so that's like a kind of experience that start starts to um be revealing so that's kind of interesting yeah it also really H understand something and really emphasized today too what you said about di really you know with their behavior actively changing what we do perceive um is that in line with learning also always comes this notion of things being Dynamic and basically due to technology technological Advance we finally at that stage where Aesthetics can hopefully also move away from this okay we show you athetic picture and then we're taking this is like the final say right this picture this it's this response and we're done with it but really looking at these things as Dynamic processes right even looking at One Singular image this is not a one and done thing your impressions change over time because your eyes move because your understanding changes um because that understanding changes possibly where you're looking so it also changes what you're going to see Etc um and that to me is a much more exciting and interesting way of of looking at these topics right it's really this Dynamic unfolding process so yeah thanks for mentioning that I think it's really important cool and also I feel like it puts action kicking off the process like I'm thinking of a food tasting like food is put into your mouth and you can report on it whereas if you have the menu then the action kicks off something that is later the aesthetic exper you choose which museum to go to you choose where to look and so that kind of front loads this action elements whereas in the contrived setting the action elements may be controlled by the experimenter and the action that the person took was to sign up for a study which is still a kind of action but then that doesn't like builds the kind of of Arc that we actually like you said paid money and take time out of life to to enjoy cool anything else you'd like to add um I would

like to add thank you um for having me um said that very briefly in the beginning but I think it's really cool to have these conversations um I think it's also nice to have it out there in the open and I hope that I didn't use all too much charagon um and I would also like to mention that uh people can very easily Google me by my name and you should be very free to get in touch with me if you're interested in any of these things I'm sometimes busy but always interested in talking with people uh no matter what they where they're coming from awesome thank you again see you next time thank you for